

Chapter 2

组合数学

Section 2.5 极值集合论

- 向日葵引理
- Erdős-Ko-Rado定理
- Sperner定理

Sunflowers

- Set system (family): $\mathcal{F} \subseteq 2^{[n]}$ with ground set $[n]$
- **Definition:** $\mathcal{F} \subseteq 2^{[n]}$ is a sunflower of **size r** with **center $\mathcal{C}$** (can be Φ)

$$\begin{cases} |\mathcal{F}| = r \\ \forall S, T \in \mathcal{F}: S \cap T = \mathcal{C} \end{cases}$$

- **Sunflower Lemma** (Erdős-Rado, 1960):

$\mathcal{F} \subseteq \binom{[n]}{k}$, when $|\mathcal{F}| > k! (r - 1)^k$, then there exists a sunflower

$$\mathcal{G} \subseteq \mathcal{F}, \text{ s.t. } |\mathcal{G}| = r$$

Proof of Sunflower Lemma

Use induction (r is given, induction on k)

1. **Basis:** $k = 1$, $\mathcal{F} \subseteq \binom{[n]}{1}$, $|\mathcal{F}| > r - 1$

$\exists r$ disjoint singletons

2. $k \geq 2$

Take **largest** $\mathcal{G} \subseteq \mathcal{F}$ with disjoint members, s.t. $\forall S, T \in \mathcal{G}$ that $S \neq T, S \cap T = \emptyset$

Case 1: $|\mathcal{G}| \geq r$. $\mathcal{G}$ is a sunflower of size r

Case 2: $|\mathcal{G}| \leq r - 1$. **Goal:** find a ‘popular’ $x \in [n]$

$\{S \in \mathcal{F} \mid x \in S\} \rightarrow (\text{remove } x) \quad \mathcal{H} = \{S \setminus \{x\} \mid S \in \mathcal{F} \wedge x \in S\}$

then $\mathcal{H} \subseteq \binom{[n]}{k-1}$. If $|\mathcal{H}| > (k-1)! (r-1)^{k-1}$, then we can use induction

Proof of Sunflower Lemma

If $\mathcal{H}$ contains a sunflower of size r , then $\{S \cup \{x\} | S \in \mathcal{H}\} \subseteq \mathcal{F}$ contains a sunflower
 $\Rightarrow \mathcal{F}$ contains a sunflower

$\mathcal{G}$: maximal matching

Take $Y = \bigcup_{S \in \mathcal{G}} S \Rightarrow |Y| \leq k(r-1)$, **claim**: $\forall S \in \mathcal{F}, Y \cap S \neq \emptyset$

If otherwise: $\exists T \in \mathcal{F}: Y \cap T = \emptyset \Rightarrow \forall S \in \mathcal{G}: S \cap T = \emptyset$ **Contradiction!**

$\Rightarrow \exists x \in Y$: # of $S \in \mathcal{F}$ containing x : $|\{S \in \mathcal{F} | x \in S\}| \geq \frac{|\mathcal{F}|}{|Y|}$

$|\mathcal{F}| > k! (r-1)^k, |Y| \leq k(r-1) \Rightarrow |\{S \in \mathcal{F} | x \in S\}| \geq (k-1)! (r-1)^{k-1}$

Erdős-Ko-Rado Theorem

➤ Sunflower Conjecture:

If $\mathcal{F} \subseteq \binom{[n]}{k}$, $|\mathcal{F}| > C(r)^k$, then $\exists$ a sunflower $\mathcal{G} \subseteq \mathcal{F}$, s.t. $|\mathcal{G}| = r$.

➤ Intersecting Family: $\forall S, T \in \mathcal{F}, S \cap T \neq \emptyset$ (trivially holds if $n < 2k$)

➤ **Erdős-Ko-Rado Theorem**

$$\mathcal{F} \subseteq \binom{[n]}{k}, n \geq 2k, \forall S, T \in \mathcal{F}, S \cap T \neq \emptyset \Rightarrow |\mathcal{F}| \leq \binom{n-1}{k-1}$$

➤ **Proof:** (induction on n and k)

Let $\mathcal{F}_0 = \{S \in \mathcal{F} | n \notin S\}$, $\mathcal{F}_1 = \{S \in \mathcal{F} | n \in S\}$,

$$1) \mathcal{F}_0 \subseteq \binom{[n-1]}{k}, S, T \in \mathcal{F}_0, S \cap T \neq \emptyset \Rightarrow \text{I. H.: } |\mathcal{F}_0| \leq \binom{n-2}{k-1}$$

$$2) \mathcal{F}'_1 = \{S \setminus [n] | S \in \mathcal{F}_1\} \subseteq \binom{[n-1]}{k-1} \text{ If } \mathcal{F}'_1 \text{ intersecting} \Rightarrow |\mathcal{F}_1| \leq \binom{n-2}{k-2}$$

Proof of Erdős-Ko-Rado Theorem

$$\Rightarrow |\mathcal{F}| = |\mathcal{F}_0| + |\mathcal{F}_1| = |\mathcal{F}_0| + |\mathcal{F}'_1| \leq \binom{n-2}{k-1} + \binom{n-2}{k-2} = \binom{n-1}{k-1}$$

Key: $\mathcal{F}'_1$ intersecting

by shifting (compression)

Goal: find a special $\mathcal{F} \subseteq \binom{[n]}{k}$, s.t. $\mathcal{F}$ remains intersecting after deleting n

➤ **Definition (Shift Operator)** $\mathcal{F} \subseteq 2^{[n]}$ for $1 \leq i < j \leq n$,

$$\forall T \in \mathcal{F}: T_{ij} = (T \setminus \{j\}) \cup \{i\} \quad (i, j)\text{-shift: } S_{ij}(\cdot)$$

$$\forall T \in \mathcal{F}: S_{ij}(T) = \begin{cases} T_{ij} & \text{if } j \in T, i \notin T, T_{ij} \notin \mathcal{F} \\ T & \text{otherwise} \end{cases}, \quad S_{ij}(\mathcal{F}) = \{S_{ij}(T) | T \in \mathcal{F}\}$$

Proof of Erdős-Ko-Rado Theorem

Proposition: 1) $|S_{ij}(T)| = |T|$, $|S_{ij}(\mathcal{F})| = |\mathcal{F}|$;

2) If $\mathcal{F}$ is intersecting, then $S_{ij}(\mathcal{F})$ is intersecting

Proof: $A, B \in \mathcal{F}$, $A \cap B \neq \emptyset$

if $j \notin A \cap B$ safe

if $j \in A \cap B$ (1) A successful, B successful; (2) A failed, B failed;

(3) A failed, B successful **Unsafe case** ($i \notin B$, $B_{ij} \notin \mathcal{F}$)

$$\textcircled{1} \quad i \in A: S_{ij}(A) = A, S_{ij}(B) = B_{ij}, S_{ij}(A) \cap S_{ij}(B) = A \cap B_{ij} = \{i\} \neq \emptyset$$

$$\textcircled{2} \quad A_{ij} \in \mathcal{F}, i \notin A, S_{ij}(A) \cap S_{ij}(B) = A \cap B_{ij} = \emptyset$$

$A_{ij} \cap B = \emptyset \Rightarrow \mathcal{F}$ is intersecting, contradiction

$\Rightarrow S_{ij}(\mathcal{F})$ is intersecting

Proof of Erdős-Ko-Rado Theorem

Repeatedly apply (i, j) -shifting $S_{ij}(\mathcal{F})$ for $1 \leq i < j \leq n$

Eventually: $\mathcal{F} = S_{ij}(\mathcal{F}), \forall 1 \leq i < j \leq n$ ‘fixed point’ $\Rightarrow \mathcal{F}$ is shifted

Assume $\mathcal{F}$ is shifted

When $n = 2k$: $\forall S \in \binom{[n]}{k}, S$ and $[n] \setminus S$ at most one in $\mathcal{F} \Rightarrow |\mathcal{F}| \leq \binom{n-1}{k-1}$

When $n \geq 2k$: arbitrary intersecting $\mathcal{F} \rightarrow$ (shifting) shifted $\mathcal{F}'$

$$|\mathcal{F}'| \leq \binom{n-1}{k-1} \Rightarrow |\mathcal{F}| \leq \binom{n-1}{k-1} \quad \mathcal{F}'_1 = \{S \setminus [n] \mid S \in \mathcal{F}_1\}$$

Now: $\mathcal{F}'_1$ is intersecting

Proof of Erdős-Ko-Rado Theorem

Assume $\mathcal{F}'_1$ is not intersecting, then $\exists A, B \in \mathcal{F}$, s.t. $A \cap B = \{n\}$

$\Rightarrow |A \cup B| = 2k - 1 < n - 1, \exists i, 1 \leq i \leq n, \text{ s.t. } i \notin A \cup B$

$C = A_{in} = (A \setminus \{n\}) \cup \{i\} \in \mathcal{F}$

$A_{in} \cap B = \emptyset \Rightarrow \mathcal{F}$ is not intersecting

Contradiction!

Katona's Proof

π : cyclic permutation, totally $(n - 1)!$

Define **k -arc**: $\mathcal{G}_\pi = \left\{ \{ \pi_{(i+j) \bmod n} | j \in [k] \} | i \in [n] \right\}$

Lemma $n \geq 2k, \forall \pi, |\mathcal{G}_\pi \cap \mathcal{F}| \leq k$

Proof: $A_i = \{ \pi_{(i+j) \bmod n} | j \in [k] \}, \mathcal{G}_\pi = \{ A_i | i \in [n] \}$

Suppose $A_t \in \mathcal{F}$, $\mathcal{F}$ is intersecting

If $A_i \in \mathcal{F}$: $A_i \cap A_t \neq \emptyset \quad t - (k - 1) \leq i \leq t + (k - 1)$

$A_i \cap A_{i+k} = \emptyset$ ($n \geq 2k$) $(k - 1)$ pairs $\{A_i, A_{i+k}\}$, $\mathcal{F}$ contains at most one of them

$$|\mathcal{G}_\pi \cap \mathcal{F}| \leq (k - 1) + 1 = k$$

Katona's Proof

Then we can use **double counting**

$$R = \{(S, \pi) | \pi \text{ is a cyclic permutation, } S \in \mathcal{G}_\pi \cap \mathcal{F}\}$$

Method 1: $|R| \leq (n-1)! \cdot k$

Method 2: $\forall S \in \mathcal{F}$: S is **continuous** under π (S is a k -arc)

$$\#: k! (n-k)! \Rightarrow |R| = |\mathcal{F}| \cdot k! (n-k)!$$

$$\Rightarrow |\mathcal{F}| \leq \frac{(n-1)!}{(k-1)! (n-k)!} = \binom{n-1}{k-1}$$

Sperner System

- $\mathcal{F} \subseteq 2^{[n]}$ POSET
- **Chain:** $S_1 \subseteq S_2 \subseteq \dots \subseteq S_k$
- **Antichain:** $\forall S, T \subseteq \mathcal{F}, S \neq T: S \not\subseteq T$
- k -uniform family: $\binom{[n]}{k}$ is an antichain
- **Q: How large can $\mathcal{F}$ be?**
- **Theorem (Sperner 1928)** $\mathcal{F} \subseteq 2^{[n]}$ If $\mathcal{F}$ is an antichain, then $|\mathcal{F}| \leq \binom{n}{\lfloor \frac{n}{2} \rfloor}$

Sperner System

➤ $\mathcal{F} \subseteq \binom{[n]}{k}$ k -uniform family

➤ **Shade** $\nabla\mathcal{F} = \{T \in \binom{[n]}{k+1} \mid \exists S \in \mathcal{F}, \text{ s.t. } S \subseteq T\}$

➤ **Shadow** $\Delta\mathcal{F} = \{T \in \binom{[n]}{k-1} \mid \exists S \in \mathcal{F}, \text{ s.t. } T \subseteq S\}$

➤ **Lemma (Sperner)** $|\nabla\mathcal{F}| \geq \frac{n-k}{k+1} |\mathcal{F}|, |\Delta\mathcal{F}| \geq \frac{k}{n-k+1} |\mathcal{F}|$

Proof: $R = \{(S, T) \mid S \in \mathcal{F}, T \in \nabla\mathcal{F}, S \subseteq T\}$

1) For any $S \in \mathcal{F}$: $n - k$ $T \in \nabla\mathcal{F}$ s.t. $S \subseteq T \implies |R| = |\mathcal{F}|(n - k)$

2) For any $T \in \mathcal{F}$: $\leq (k + 1)$ $S \in \mathcal{F}$ s.t. $S \subseteq T \implies |R| \leq |\nabla\mathcal{F}|(k + 1)$

$\implies$ **Double Counting:** $|\nabla\mathcal{F}| \geq \frac{n-k}{k+1} |\mathcal{F}| \implies$ When $k \leq \frac{n-1}{2}$, $|\nabla\mathcal{F}| \geq |\mathcal{F}|$

Sperner System

➤ **Proposition:** $\mathcal{F} \subseteq 2^{[n]}$, $\mathcal{F}_k = \mathcal{F} \cap \binom{[n]}{k}$, let $k_{\min}$ be the smallest k such that $|\mathcal{F}_k| > 0$, $k_{\max}$ be the largest k such that $|\mathcal{F}_k| > 0$

$$\mathcal{F}' = \begin{cases} \mathcal{F} \setminus \mathcal{F}_{k_{\min}} \cap \nabla \mathcal{F}_{k_{\min}} & \text{if } k_{\min} \leq \frac{n-1}{2} \\ \mathcal{F} & \text{otherwise} \end{cases}, \mathcal{F}'' = \begin{cases} \mathcal{F} \setminus \mathcal{F}_{k_{\max}} \cap \Delta \mathcal{F}_{k_{\max}} & \text{if } k_{\max} \geq \frac{n+1}{2} \\ \mathcal{F} & \text{otherwise} \end{cases}$$

$\Rightarrow |\mathcal{F}'| \geq |\mathcal{F}|, |\mathcal{F}''| \geq |\mathcal{F}|$; **If $\mathcal{F}$ is an antichain, then $\mathcal{F}', \mathcal{F}''$ are also antichains**

➤ **Proof:** By contradiction. If $\mathcal{F}'$ is NOT an antichain

$\Rightarrow \exists S, T \in \mathcal{F}'$, s.t. $S \subseteq T$ One of S, T must be in $\nabla \mathcal{F}_{k_{\min}}$ (**S**; $T \in \mathcal{F}$)

$\Rightarrow \exists R \in \mathcal{F}_{k_{\min}} \subseteq \mathcal{F}$, s.t. $R \subseteq S \subseteq T$ Contradiction! ($\mathcal{F}$ is an antichain)

Proof of Sperner's Theorem

➤ $\mathcal{F}_k = \mathcal{F} \cap \binom{[n]}{k} = \{S \in \mathcal{F} \mid |S| = k\}$

- for the smallest k that $|\mathcal{F}_k| > 0$ and $k \leq \frac{n-1}{2} \Rightarrow$ replace $\mathcal{F}_k$ by $\nabla \mathcal{F}_k$
- for the largest k that $|\mathcal{F}_k| > 0$ and $k \geq \frac{n+1}{2} \Rightarrow$ replace $\mathcal{F}_k$ by $\Delta \mathcal{F}_k$

$$\Rightarrow \mathcal{F} \subseteq \binom{[n]}{\lfloor \frac{n}{2} \rfloor} \Rightarrow |\mathcal{F}| \leq \binom{n}{\lfloor \frac{n}{2} \rfloor}$$

Another Proof by Lubell

➤ π : permutation of $[n]$ $S \subseteq [n]$ **prefixes** π : $S = \{\pi_1, \pi_2, \dots, \pi_{|S|}\}$

➤ Given S , # of π prefixed by S : $|S|! (n - |S|)!$

$\mathcal{F}$ is an antichain: $\forall S, T \in \mathcal{F}, S \not\subseteq T \Rightarrow$ no permutation π can be prefixed by ≥ 2 members in $\mathcal{F}$

➤ $\sum_{S \in \mathcal{F}} |S|! (n - |S|)! \leq n! \Rightarrow \sum_{S \in \mathcal{F}} \frac{|S|!(n - |S|)!}{n!} = \sum_{S \in \mathcal{F}} \frac{1}{\binom{n}{|S|}} \leq 1$ (**LYM inequality**)

➤ $\binom{n}{|S|} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor} \Rightarrow \sum_{S \in \mathcal{F}} \frac{1}{\binom{n}{|S|}} \leq |\mathcal{F}| \frac{1}{\binom{n}{\lfloor \frac{n}{2} \rfloor}} \leq 1 \Rightarrow |\mathcal{F}| \leq \binom{n}{\lfloor \frac{n}{2} \rfloor}$

LYM inequality

➤ **Lubell, Yamamoto, Maschalkin: LYM inequality**

$\mathcal{F} \subseteq 2^{[n]}$ is an antichain, then $\sum_{S \in \mathcal{F}} \frac{1}{\binom{n}{|S|}} \leq 1$

➤ **Proof: (Alon) Probabilistic Method**

π : uniform random permutation on $[n]$

$\mathcal{C}_\pi = \{ \{\pi_1\}, \{\pi_1, \pi_2\}, \dots, \{\pi_1, \pi_2, \dots, \pi_n\} \}$ (maximal chain)

$\forall S \in \mathcal{F}, \chi_S = \mathbf{1}\{S \in \mathcal{C}_\pi\} \Rightarrow \mathbb{E}[\chi_S] = \Pr(S \in \mathcal{C}_\pi) = \frac{1}{\binom{n}{|S|}}$ (uniform)

$X = \sum_{S \in \mathcal{F}} \chi_S \Rightarrow \mathbb{E}[X] = \sum_{S \in \mathcal{F}} \frac{1}{\binom{n}{|S|}}, X = |\mathcal{F} \cap \mathcal{C}_\pi| \leq 1$ (antichain)